

MPG No. 13.1.44 $4mm$ [Type I, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxxy} & -M_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & M_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & -M_{xxyx} & 0 & 0 & -M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}$$

$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{yzzx} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxy} & 0 \\ -M_{xxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xxy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxy} = \mathbb{M}_{4z}^{\alpha(1)}$$